

creases with increase in temperature.

At room temperature, transistor T2 conducts. When temperature of the sensor goes above 70°C [depending on thermistor constant (K)], T2

stops conducting. When T2 stops conducting, an alert signal is generated in RPI, which is sent as a push message to the Android phone.

Most doorbells produce pulse-wave modulation waves

while ringing. You need to hack the doorbell so that it produces at least 2V (pulse-wave modulation with low-pass filter) output signal. You may use this signal as an interrupt signal to drive RPi pin and generate an alarm log in your Android phone. Door-bell signal is fed to RPi through CON4.

When someone rings the doorbell, you will get a notification on your phone. Various alert notifications received on Android smartphone are shown in Fig. 3.

Software

Pushover makes it easy to get real-time notifications on your smartphone. Follow the steps given below to activate Pushover app on your Android device.

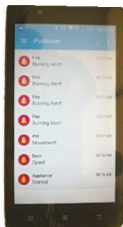


Fig. 3: Alert notifications in Android mobile

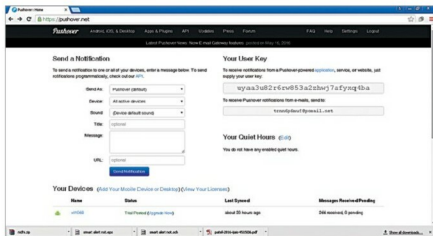


Fig. 4: Typical user key generated on Pushover app

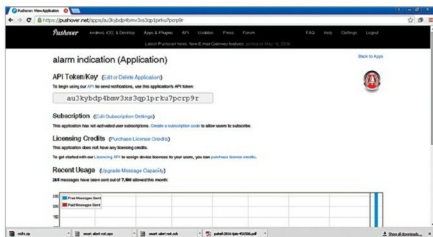


Fig. 5: Typical API key

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

1. On your mobile device, open Pushover app and log into your account. You will be asked to register your device by providing a name for it.

2. Visit <https://pushover.net> on your computer and log into your account. You should see your devices listed in a section called Your Devices, a little way down the page. Verify that the devices you have registered are on this list and are enabled.

3. Note down your user key (like, uyaa3u82r6w853a2zhw7afyxq4ba) as shown in Fig. 4. You will need it for your Python code.

4. To get API key, click on Your Applications (doorbell) and note down your API token/key (like, au3kybdp4bmv3x3qp1prku7pcrp9r) as shown in Fig. 5.

5. Under Settings in Pushover, you can set an audio alert for quiet hours, etc for your mobile as per your requirement.

Testing Pushover app

Open Pushover app on your Android mobile. Open appliance.py file on the terminal in RPi. Change user_key and application_key in the code with your keys obtained as shown in Figs 3 and 4. Run the following command:

```
$ sudo python appliance.py
```

With the server connected to RPi, you will get 'Appliance Server Started' message in the terminal and 'Appliance Started' message on your mobile. As per the interrupt signal received from the input sensors (motion, door break, fire, doorbell, etc), the corresponding alert notification will be received on your Android phone through Pushover app. **EFY**



Biswajit Das was manager - R&D, EFY Labs, when he prepared this article